

FBD parameters

M = mass of body

m = mass of each wheel

L_{cg} = Distance from wheel axis to center of mass

r = wheel radius

I_{cg} = Moment of Inertia of body

J = Moment of inertia of each wheel

g = gravity

$2T_w$ = Total motor torque applied to wheels

Mathematical Model (2DOF Self balancing robot)

Model the robot as an inverted pendulum mounted on two wheels

Coordinates

x is horizontal displacement

θ is the tilt angle from vertical

ϕ wheel rotation

Centre of Mass (COM)

$$x_c = x + L \sin \theta$$

$$y_c = L \cos \theta$$

Velocity of COM

we differentiate

$$\dot{x}_c = \dot{x} + L \dot{\theta} \cos \theta$$

$$\dot{y}_c = -L \dot{\theta} \sin \theta$$

Acceleration of COM

we differentiate

$$\ddot{x}_c = \ddot{x} + L \ddot{\theta} \cos \theta - L \dot{\theta}^2 \sin \theta$$

$$\ddot{y}_c = -L \ddot{\theta} \sin \theta - L \dot{\theta}^2 \cos \theta$$

we are using Newton's Second Law ($F = ma$)
For x direction

$$\sum F_x = M \ddot{x}_c$$

Forces

Friction Force

$$F_r = M(\ddot{x} + l\ddot{\theta}\cos\theta - l\dot{\theta}^2\sin\theta)$$

For wheels

$$(M+2m)\ddot{x} + Ml\ddot{\theta}\cos\theta - Ml\dot{\theta}^2\sin\theta = F_r$$

Rotation at body at COM

$$\sum \tau_{CG} = I_{CG}\ddot{\theta}$$

$$g = Mgl\sin\theta$$

$$\therefore I_{CG}\ddot{\theta} = Mgl\sin\theta - Ml\ddot{x}\cos\theta$$

$$I_{CG}\ddot{\theta} + Ml\ddot{x}\cos\theta = Mgl\sin\theta$$

Rotation of wheels

$$\sum \tau = 2J\ddot{\phi}$$

$$2J\ddot{\phi} = 2T_w - F_r r$$

Rolling with no slip condition

$$\ddot{x} = r\ddot{\phi}, \quad \ddot{\phi} = \frac{\ddot{x}}{r}$$

From wheel equation

$$F_r = \frac{2T\omega - 2J\dot{\phi}^{\circ\circ}}{r}$$

$$\dot{\phi}^{\circ\circ} = \frac{\ddot{x}}{r}$$

$$F_r = \frac{2T\omega}{r} - \frac{2J}{r^2} \ddot{x}$$

Substitute in Newtons Second Equation

$$(M+2m)\ddot{x} + M\ddot{\theta}\cos\theta - M L \dot{\theta}^2 \sin\theta = \frac{2T\omega}{r} - \frac{2J}{r^2} \ddot{x}$$

$$\therefore \left(M+2m + \frac{2J}{r^2} \right) \ddot{x} + M\ddot{\theta}\cos\theta - M L \dot{\theta}^2 \sin\theta = \frac{2T\omega}{r}$$

The equations of Motions

Rotation

$$I_{CG} \ddot{\theta} + M L \ddot{x} \cos\theta = M g l \sin\theta$$

Translation or newton's second law equation

$$\left(M+2m + \frac{2J}{r^2} \right) \ddot{x} + M\ddot{\theta}\cos\theta - M L \dot{\theta}^2 \sin\theta = \frac{2T\omega}{r}$$

$$\frac{2T\omega}{r} = \text{motor force}$$

$$\frac{2J}{r^2} = \text{wheel inertia}$$

$$Ml\ddot{\theta}\cos\theta = \text{coupling}$$

$$Mgl\sin\theta = \text{gravity}$$

$$Ml\ddot{\theta}\sin\theta = \text{centrifugal force}$$